Azure cloud, Azure DevOps, Any Microsoft centric tools experience: Power Platform, MSFabric Admins etc... completely MS centric focused

i need interview questions and answrs

Memory updated

Here's a set of **Microsoft-centric interview questions and answers** categorized by topic: **Azure Cloud, Azure DevOps, Power Platform, and MS Fabric Admin.**

🔵 Azure Cloud

Q1. What are Azure Resource Groups and why are they important?
A: Resource Groups in Azure are containers that hold related resources for an Azure solution. They help manage and organize resources effectively, enable role-based access control (RBAC), cost tracking, and resource lifecycle management.

Q2. Explain the difference between Azure Availability Sets and Availability Zones.
A:

- **Availability Sets** protect against hardware failures within a data center.
- **Availability Zones** are physically separate data centers within a region, providing better redundancy and fault isolation.

Q3. How do you secure an Azure Storage Account?
A:

- Use **Private Endpoints** or service endpoints
- Enable **soft delete** and **immutable blobs**
- Use **Azure RBAC** and **Shared Access Signatures (SAS)**
- Enforce **TLS** and enable **firewall rules**

🟣 Azure DevOps

Q4. How do you structure a multi-stage YAML pipeline in Azure DevOps?
A: A multi-stage pipeline defines stages like `Build`, `Test`, `Deploy`. Each stage has its own jobs and steps. Example:

```yaml
stages:
- stage: Build
  jobs:
  - job: BuildApp
    steps:
    - script: dotnet build

- stage: Deploy
  dependsOn: Build
  jobs:
  - job: DeployToAzure
    steps:
    - task: AzureWebApp@1
```

Q5. How do you manage secrets in Azure DevOps pipelines?
A:

- Use **Azure Key Vault** with `AzureKeyVault@2` task
- Define secrets in **Pipeline Library (Variable Groups)** and mark them as `secret`
- Avoid using secrets in plain text; use environment variables instead

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🟢 Power Platform (Power Automate, Power Apps, Power BI)

Q6. What is the Common Data Service (Dataverse)?
A: Dataverse is a secure and scalable data platform used by Power Platform apps. It standardizes data and business logic across applications.

Q7. How would you integrate Power Automate with Azure DevOps?
A:

- Use built-in connectors in Power Automate for Azure DevOps
- Automate tasks like work item creation, approvals, or notifications on pipeline events
- Use webhooks and custom connectors for advanced integrations

Q8. How do you manage security in Power Apps?
A:

- Control access via **Azure AD roles**, **Dataverse security roles**, and **sharing settings**
- Implement **row-level security**
- Use **environment-level** and **app-level** permissions

🟠 Microsoft Fabric Admin (MSFabric)

Q9. What is Microsoft Fabric (OneLake) and its purpose?
A: Microsoft Fabric is an end-to-end analytics platform that unifies data engineering, integration, warehousing, real-time analytics, and BI using OneLake—a unified data lake.

Q10. What is your role as an MS Fabric Admin?
A:

- Manage **workspaces, capacity, and licenses**
- Configure **OneLake security and data lineage**
- Monitor and optimize **dataflows, pipelines, and workloads**
- Ensure **compliance, governance**, and **data access policies**

Would you like these in a **PDF or table format**? Or do you want **more advanced scenario-based questions** per topic?

more advanced scenario-based questions per topic?

Great! Here's a **more advanced, scenario-based set of interview questions and answers** for the Microsoft-centric stack: **Azure Cloud, Azure DevOps, Power Platform, and MS Fabric Admin.**

🔵 Azure Cloud – Scenario-Based Questions

Q1. Your app must handle traffic spikes and maintain uptime. How would you design the infrastructure?
A:

- Use **Azure Load Balancer or Application Gateway** with **Autoscale-enabled VM Scale Sets or App Service Plans**.
- Deploy across **Availability Zones**.
- Use **Azure Front Door** or **Traffic Manager** for global traffic distribution and failover.
- Integrate with **Azure Monitor** and **Log Analytics** for auto-healing and alerts.

Q2. How would you implement secure access to a multi-tier application hosted in Azure?
A:

- Place tiers in different **subnets**, use **NSGs** to restrict access.
- Use **Private Endpoints** for backend services.
- Leverage **Managed Identities** for internal access without secrets.
- Enforce **Just-in-Time VM access** and **conditional access policies**.

🟣 Azure DevOps – Scenario-Based Questions

Q3. You need to deploy infrastructure and application code in a single pipeline. How do you structure it?
A:

- Use a **multi-stage YAML pipeline** with separate `Infra` and `App` stages.
- Infra stage uses **Terraform/Bicep with** `TerraformCli@1`, **task or** `AzureCli@2`.
- App stage uses **build & deploy tasks** (e.g., `DotNetCoreCli`, `AzureWebApp`)
- Use **pipeline artifacts** to pass values like IPs or connection strings.
- Example:

```yaml
stages:
- stage: Infra
  jobs:
    - job: DeployInfra
      steps:
        - task: TerraformCli@0

- stage: App
  dependsOn: Infra
  jobs:
    - job: DeployApp
      steps:
        - task: AzureWebApp@1
```

Q4. A team complains that secrets are leaking via logs in the pipeline. How do you prevent this?
A:

- Use **Azure Key Vault** and link secrets using KeyVault task.
- Ensure sensitive variables are marked as `secret` in YAML or UI.
- Add `echo "***"` for secret-related outputs.
- Avoid logging `$(secretVar)` directly. Use secure logging methods.

🟢 Power Platform – Scenario-Based Questions

Q5. A business wants to automate approval for document uploads to SharePoint with notifications. What's your solution?
A:

- Use **Power Automate** with triggers: `When a file is created in SharePoint`
- Add **approval flow** using `Start and wait for an approval`
- Based on outcome, send email (Outlook connector) or move file
- Add conditions for escalation and error handling (parallel branches)

Q6. You need to control access to sensitive Power BI reports shared across departments. How would you enforce security?
A:

- Use **row-level security (RLS)** in the dataset
- Implement **security groups** in Azure AD and bind to RLS roles
- Publish reports in a **dedicated workspace**
- Use **Power BI Service access controls** and disable sharing/export options

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🟠 Microsoft Fabric Admin (MSFabric) – Scenario-Based Questions

Q7. Users are complaining of slow performance in Lakehouse queries. What steps do you take to diagnose and resolve?
A:

- Review **Fabric workspace capacity metrics**
- Analyze **Dataflows and pipeline performance**
- Check if **OneLake file format (Delta, Parquet)** is optimized
- Validate **Partitioning, Caching, and Indexing** settings
- Scale up or redistribute compute resources using **capacity management tools**

Q8. You are asked to govern data access across multiple Fabric projects. How would you implement it?
A:

- Use **Microsoft Purview** for data governance and access policies
- Assign **roles via RBAC** in Fabric Admin portal
- Use **workspace-based access**, limit access to OneLake directories
- Enforce **data lineage and sensitivity labels**
- Enable **audit logs** to track access and changes

Want to go deeper into any of these (e.g., Power Automate advanced integrations, Azure DevOps testing, or MS Fabric data pipelines)? Or should I generate a downloadable interview doc from this set?